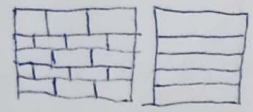
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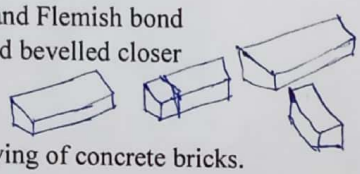
**UNIVERSITY OF ILORIN**

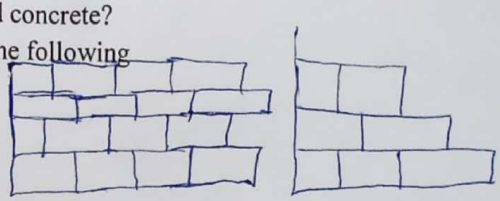
**DEPARTMENT OF CIVIL ENGINEERING (SWEP 2018) (Answer all) TIME :1HOUR**

1. (a) Why is reinforced concrete regarded as composite material?
   (b) What are the major functions of reinforcement in reinforced concrete?
2. With the aid of a well labeled diagram. Differentiate between the following

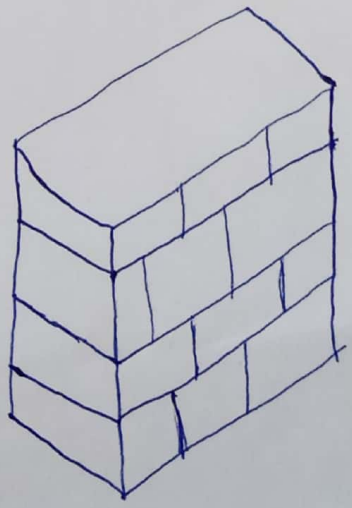


   (v) Toothing and racking back
   (vi) Stretcher bond and Flemish bond
   (vii) Queen closer and bevelled closer
   (viii) Bat and ¾ bat.





3. List five (5) tools used in laying of concrete bricks.
   (b) Highlight briefly the uses of the tools mentioned in (3a) above
4. Differentiate between the following:
   (vi) Brick and block
   (vii) Mortar and all –in – aggregates
   (viii) Lime mortar and concrete
   (ix) Guaged mortar and cement mortar
   (x) Falling out and blowing



droughtness

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...functions of reinforcement in reinforced concrete?

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UNIVERSITY OF ILORIN
FACULTY OF ENGINEERING AND TECHNOLOGY
THIRD SEMESTER EXAMINATIONS 2017/2018 SESSION
ABE 222/BME222/CHE 222/CVE 222/CPE 222/ELE 222/MEE 222/FBE 222/MME 222/WRE 222
2017/2018 SWEP SPECIAL SKILL ACQUISITION EXAMINATION
MATERIALS AND METALLURGICAL ENGINEERING

Time Allowed: 2 Hours

INSTRUCTION: Attempt all questions.

1. A taper provided on the pattern for its easy and clean withdrawal from the mould is known as ----------.
   a) distortion allowance
   b) shrinkage allowance
   c) draft allowance
   d) machining allowance
2. In order to deliver molten metal from pouring basin to gate,
   a) a riser is used
   b) a core is used
   c) a sprue is used
   d) a gagger is used
3. A sand employed on the faces of the pattern before moulding is called ----------
   a) green sand
   b) dry sand
   c) parting sand
   d) oil sand
4. The purpose of a riser is to
   a) deliver molten metal into the mould cavity
   b) deliver molten metal from pouring basin to gate
   c) act as a reservoir for the molten metal
   d) feed the molten metal to the casting in order to compensate for the shrinkage
5. Cores are used to
   a) form internal cavities in the casting
   b) improve mould surface
   c) form a part of green sand mould
   d) all of the above
6. The property of sand due to which the sand grains stick together is called
   a) collapsibility
   b) permeability
   c) cohesiveness
   d) adhesiveness
7. When the molten metal is fed in the cavity of a metallic mould by gravity, the method of casting is known as ---------.
   a) permanent mould casting method
   b) die casting method
   c) centrifugal casting method
   d) investment casting method
8. The region added to a pattern to locate and support a core is ……
   (a) Core seat (b) Core shift (c) Core gate (d) Core print
9. A casting defect which results in the general enlargement of a casting is known as
   a) shift
   b) swell
   c) sand wash
   d) blow hole
10. A casting defect which occurs near the in-gates as rough lumps on the surface of a casting is known as
    a) Scab
    b) Swell
    c) Sand wash
    d) Shift
11. The ability of moulding sand to allow for easy passage of steam/gases is known as
    a) Porosabilty
    b) Flowability
    c) Permeability
    d) Poroability
12. The ability of the moulding sand to withstand high temperature is referred to as its
    a) hardness
    b) hotness
    c) refractoriness
    d) stiffness
13. The purpose of a gating system is to
    a) deliver molten metal into the mould cavity
    b) act as a reservoir for the molten metal
    c) feed the molten metal to the casting in order to compensate for the shrinkage
    d) deliver molten metal from pouring basin to gate
14. …… is added to the mould to create a parting line.
    (a) Parting powder (b) Sodium silicate (c) Bentonite (d) kaolin

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15 The region added to a pattern to locate and support a core is .....

(a) Core seat (b) Core shift (c) Core gate (d) Core print

16 .....serves as a slag-trap.

(a) Core (b) Runner bar. (c) Riser (d) Core print

17 Sodium silicate may be used as a ..... in the Foundry.

(a) Glassy agent (b) Flux (c) Binder (d) Core

18 .....is a possible defect in casting

19 ----------, ---------- and ---------- are type of casting process

20. A.....is a rigid body with a cavity used to give shape to molten metal as it solidifies (a) Riser (b) Mould (c) Cope (d) Core

21. The provision made on the pattern to compensate for dimensional changes that occur during casting is .....

(a) Core print (b) Allowance (c) Space (d) Addition

22 The bottom part of a mould box is referred to as.....

(a) Core (b) Cope (c) Riser (d) Drag

23 .....is the ability of the molding sand to easily worked.

(a) Fluidity (b) Floatability (c) Flowability (d) Adhesiveness

24. A type of Pattern is .....(a) Separated pattern

(b) Combination (c) Consolidated pattern (d) Split pattern

25 An example of binder used in the foundry workshop for the preparation of a mold is.....

(a) Bentonite (b) Cement (c) Sillimanite (d) Magnesite

26 .....is an insert in the mould used for producing internal features in the casting.

27 .....is a replica of the object to be made

28. The molding sand is referred to as "**green**" sand because the sand contains..... (a) Silica (b) Moisture (c) Binder (d) Kaolin

**DEPARTMENT OF CHEMICAL ENGINEERING (SSAT)**

1. Gaseous suspension paint is always suspended in -------- or -------- form in a gas that is sprayed on an object

2. Liquid paint application is applied directly using the following

(a) brush (b) blade (c) roller (d) hand

Use this to answer question 3 to 7. To produce 27.5 liters of emulsion paint, calculate the amount of the following

3. calcium carbonate required (a) 8kg (b) 9.6kg (c) 11kg

4. Titanium oxide (a) 500g (b) 1.375kg (c) 1.750 kg

5. Polyvic acrylic (a) 2 liters (b) 1.5 liter (c) 1375 ml

6. De-foamer (a) 100ml (b) 110ml (c) 137.5ml

7. Water (a) 16500ml (b) 9625ml (c) 12000ml

Calculate the amount of the following materials needed to produce 6litres of gloss paint. Use this to answer question 8 to 10.

8. Resin (a) $^1/_2$kg (b) 1kg (c) $^3/_2$ kg

9. Thickner (a) 2 kg (b) $2^1/_2$ kg (c) $1^1/_2$ kg

10. Kerosene (a) 3liters (b) 270ml (c) 240ml

11. ---------- protect the substrate from the harmful effect of ultraviolet light

12. Binder can be categorized according to ---------- or ----------

13. Drying and curing are two different processes, Yes or No

14. The main purpose of solvent is to adjust the (a) viscosity and drying (b) curing and drying (c) curing and viscosity

15. ---------- controls flow and application properties

16. ---------- function as a carrier for non-volatile components (a) binder (b) pigment (c) solvent

17. Solvents can also act as diluent in paint, therefore ---------- is the main diluent for the emulsion produced

18. ---------- additive is used to modify surface tension and increase wet edge in paint

19. ---------- is added to fight bacterial growth in paint

20. Powder coating is used for painting (a) metal surface (b) brick surface (c) none of the above

21. In an exothermic chemical reactor, the reactant has (a) higher temperature (b) more energy (c) less energy (d) less temperature

22. Parameter controlled while operating a chemical reactor include the following except (a) volume (b) molecular weight (c) temperature (d) concentration

23. From the following, choose one which is not part of chemical reactor (a) baffle (b) blade (c) stirrer (c) catalyst

24. The most suitable reactor for paint production is (a) batch reactor (b) semi batch reactor (c) fluidized bed reactor (d) Continuous Stirred Tank Reactor (CSTR)

25. The energy of reactants in a chemical reaction which transforms into product is termed as (a) potential energy (b) binding energy (c) threshold energy (d) activation energy

26. A catalyst is a substance which ---------- chemical reaction

(a) increases the speed of a (b) decreases the speed of a (c) can neither increase nor decrease the speed of a (d) alters the value of equilibrium content in a reversible

27. A rate of chemical reaction is not affected by the (a) catalyst (b) temperature (c) number of molecules (d) reactant's concentration

28. The essence of designing a chemical reactor is to maximized (a) labour cost (b) raw material cost (c) net profit value (d) contingencies

29. Which of these reactors involve steady state operation (a) batch reactor (b) semi batch reactor (c) CSTR (d) all of the above

30. Draw and label a typical chemical reactor.

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THIRD SEMESTER EXAMINATIONS 2017/2018 SESSION

ABE 222/BME222/CHE 222/CVE 222/CPE 222/ELE 222/MEE 222/FBE 222/MME 222/WRE 222

2017/2018 GENERAL QUESTIONS

AGRICULTURAL AND BIOSYSTEM ENGINEERING

**Automobile (Diesel) 15 mins**

INSRUCTION: ANSWER ALL THE QUESTIONS

1. The two main power transmission systems of the tractor are (a) Crank shaft & connecting rod (b) piston & gasket (c) PTO & Hydraulic (d) PTO & Draw bar
2. The firing order of a Four-cylinder engine is (a) 1-3-4-2 (b) 1-3-2-4 (c) 4-3-2-1 (d) 1-2-3-4

(a) A Farm implement that is connected on the drawbar of the tractor is called----------(a) Mounted implement (b) semi mounted implement (c) Drawbar implement (d) Trailed implement

3. The ----------of the CI engine is higher than that of SI engine (a) speed (b) compression ratio(c) Auto ignition temperature (d) fuel consumption
4. One of these is not associated with compression stroke(a) fuel injection (b) high air temperature (c) pressurized air (d) valves closed
5. Which of these engine strokes is used to describe the firing order? (a) induction (b) compression (c) power (d) exhaust
6. One of the reasons why gasoline may not be used in a CI engine is……(a) high auto-ignition temperature (b) high viscosity (c) high speed (d) high ductility
7. The fuel efficiency of the CI engine is because of it (a) injector system (b) carburetor system (c) carburetor system (c) hydraulic system(d) high torque
8. Which of these is a fully mounted implement? (a) tipper (b) manure spreader(c) multi-gang plough (d) lawn mower
9. One of these converts chemical energy to heat (a) alternator(b) Carburetor(c) transformer(d) engine
10. The exhaust analyser is used for (a) exhaust stroke (b) engine diagnostics (c) silencer (d) pressure analysis
11. One of these implements is powered by the tractor hydraulic system (a) tipper (b) trailers (c) tankers (d) agitators
12. One of these is not part of the routine checks done on the tractor. (a) check oil level (b) hitch the tractor frequently (c) check air of tyres (d) check elasticity of fan belt
13. Which of these is powered by the PTO shaft of the tractor. (a) rotary plough (b) disc plough (c) mould board plough (d) disc harrow
14. The general order in which an implement may be hitched to a tractor is …(a) Top-right-left (b) top-left-right (c) left-right-top(d) right-left-top
15. Servicing an engine does not entail one of the following except (a) greasing the joints (b) changing the oil (c) change the oil filters (d) changing the air cleaners
16. As a prime mover, the tractor does not entail one of the following except (a) hitch implements (b) power implements (c) move implements (d) transform implements
17. Which of these is not part of the Induction Stroke? (a) downward stroke (b) inlet valve opens (c) fuel intake (d) air intake
18. In a CI engine, ignition of fuel is achieved by (a) spark plug(b) ignition key (c) compression forces (d)injectors
19. The cam shaft does one of the followings (a) rotates the gears (b) operates the injector pump (c) opens and closes the valves (d) connects the PTO
20. Which of these components converts electrical to mechanical energy? (a) kick starter (b) alternator (c) battery (d) capacitor
21. One of these is a semi-mounted implement (a) disc plough (b) row-crop planter(c) slasher (d) yam harvester



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22. One of these receives direct power from the kick starter (a) cam shaft (b) flywheel(c) alternator (d) battery
23. The compression ring acts as a (a) sealant (b) compressor(c) lubricant (d) connector
24. To transform the reciprocating action of the piston into rotary power of the crankshaft, we need (a) piston shaft (b) oil ring (c) crank case (d) connecting rod
25. What determines the specific function of the tractor? (a) the type of fuel used (b) the type of implement attached (c) the power transmission system
26. What is the type of energy stored in a car battery? (a) chemical (b) electrical (c) heat (d) mechanical
27. Which of these is not true of compression stroke of a CI engine? (a) both valves are closed(b) the volume of air is reduced (c) gases are at atmospheric pressures (d) fuel is not yet introduced
28. The fuel system of the engine is likened to the ---------- of the human body (a) circulatory system (b) nervous system (c) respiratory system (d) digestive system
29. One of these is not found on the cylinder head of the engine (a) Piston (b) Nozzle (c) Rocker arms (d) valves springs

**Welding /Fabrication unit**

**INSTRUCTION:**

a. Fill in the blank spaces only
b. Time allowed: **15 minutes**

1. A welding transformer regulates ---------- in Arc welding
2. Spraying in fabrication comes immediately after-----, -------&--------stages.
3. ----------&-------- are not methods of joining permanently in fabrication.
4. An official writing tool in Fabrication workshop is----
5. Major types of Welding in Fabrication are ------&-----
6. A spanner will help to ----------&----------
7. The Bench vice is a device to render------in a workshop
8. The Welder is also referred to as ------&----------
9. Oxy-acetylene welding operation requires a ---------- to protect the eyes
10. Welding electrode is not a requirement of ---------- -welding
11. Welding is not applicable to ----------materials
12. The -------and the----------must be earthed in Arc welding operation
13. The Welding rod is a requirement of ----------welding
14. ---------welding can also be reffered to as gas welding
15. Marking-out procedure in Fabrication comes after----
16. In pWelding, metals are joined together ----------
17. -------, ------& ------are types of spanners in a workshop
18. Spraying in Fabrication is to ---------&----------products
19. The Guillotine shear is a ------&------equipment in Fabrication
20. Oxy-acetylene welding has ----------&----------as its major requirements
21. ----------&----------are Arc welding accessories
22. Welding shield is to Arc welding as -----is to gas welding
23. Joining methods in Fabrication are ------,------&--------
24. The Workbench gives------to the Bench Vice in a workshop
25. Engineering drawings are produced at ----------stage
26. Flame is also a ----------device in fabrication
27. "Steel rule" is a ----------measuring tool
28. ----------exists between Electrode holder and Ground or Welding cable
29. Earth clamp is an accessory of ----------welding
30. Finishing /Dressing stage requirements in a workshop are ----------&---------
31. In Arc welding -------is used for the protection of the eyes
32. Arc welding operation on a wet surface must be avoided for reason of -------
33. A precision tool measures ----------in a workshop
34. Sheet metals have ----------as their major cutting device in a Workshop
35. Internal/External Calipers are tools that ---------- measurements in construction.

**DEPARTMENT OF WATER RESOURCES AND ENVIRONMENTAL ENGINEERING** 15 MINS

(1) The following are major categories of plumbing system except?
(a) fuel gas piping
(b) rainwater, surface and sub-surface water drainage
(c) septic systems
(d) All of the above



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(2) Which of the followings best describe a plumbing system?

(a) Providing water for drinking alone

(b) Entire process of providing water supply into a building, removal of wastes and excess water out of a building.

(c) Entire process of removal of wastes and excess water out of a building.

(d) None of the above

(3) Pipes are made up of the following materials except?

(a) Zinc

(b) Copper

(c) Lead

(d) Polyvinyl Chloride

(4) One of the major advantages of using PVC pipes are

(a) Does not corrode and cheaper

(b) Readily available in the market

(c) Commonly used by the plumber

(d) Flexible and Fragile

(5) Convert 15mm Pipe to inches

(a) ½"

(b) ¼"

(c) 3/4"

(d) 1 1/2"

(6) The following are examples of plumbing fitting except?

(a) Gauge Valves (Stop Cork), Elbow and Union

(b) Tee, Plug and water closet (WC)

(c) Adaptor, Union and Elbow

(d) Adaptor, Union and Plug

(7) Which of these option is not a plumbing tool?

(a) Digger, Hack saw and Spade

(b) Adaptor, Union and Tee

(c) Yarn rope, Wrench and Hammer

(d) Spirit level, yarn rope and Hammer

(8) All the following are sanitary wares expect?

(a) Bathtub

(b) Urinals

(c) Water Closet

(d) Cistern

(9) The following are types of joint except?

(a) Union Joint

(b) Tee Joint

(c) Kay Joint

(d) Elbow Joint

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(10) What will result when a 4-inch pipe is reduced in size in the direction of flow?

(a) The waste will travel faster which will allow for better cleaning of the pipe

(b) This could cause blockage to build up at the reduction point

(c) This could cause slugging in the waste system

(d) This will have no effect on the drainage system

(11) All pipe used in plumbing applications will have all of the following markings except?

(a) Pressure rating in psi

(b) Nominal pipe size

(c) Length of pipe

(d) Manufacturer's brand name

(12) The provisions for licensing are determined by which of the following?

(a) State Board of plumbing

(b) Authority having jurisdiction

(c) Plumbing inspector

(d) The responsible master plumber

(13) Which of the following is not a merit of direct cold water supply?

(a) It is cheaper to install

(b) Growth of harmful bacteria does not occur

(c) Can deal with large demand

(d) There is emergency backup supply

(14) The following are demerit of direct cold-water supply except?

(a) There is no emergency backup supply

(b) Drop in supply occurs at low pressure

(c) Prone to more system noise (i.e Water Harmer) as water is under high pressure

(b) Growth of harmful bacteria does not occur

(15) Identify the odd in indirect cold-water supply:

(a) There is emergency backup

(b) Can supply cold water even if mains water fails

(c) Less system noise as water is not under pressure

(d) It is cheaper to install



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(16) All the following are disadvantages of indirect cold-water supply except?

(a) Expensive to install

(b) Can deal with large demand

(c) There is chance of growing harmful bacteria

(d) Cannot deal easily with large demand

(17) The following are plumbing fitting except?

(a) Elbow

(b) Bend

(c) Cesspool

(d) Adaptor

(18) What is the function of a Plunger?

(a) Cleaning the toilet system after use

(b) Cleaning the floor of the toilet

(c) Cleaning and draining foul water

(d) None of the above

(19) A sewer is an?

(a) Artificial Pipe for carrying sewage

(b) Artificial Pipe for carrying water

(c) Artificial Pipe for carrying sewerage

(b) None of the above

(20) Rain water can be disposed as follows except?

(a) Disposed into public sewer

(b) Disposed into public natural stones

(c) Disposed into public soak away

(d) Disposed into public mains

(21) The size of pipes from mains are:

(a) Equivalent to the sizes of pipes into apartments

(b) Larger than the size of pipes for the office-toilets

(c) More economical

(d) All of the above

(22) Should the ball float of the float operated valve in a flushing cistern become full of water?

(a) The cistern would continuously flush

(b) The cistern would fall to flush

(c) Water would discharge from the overflow

(d) Water hammer would be apparent

(23) When running a hot water pipe through a wall it should be sleeved to

(a) Allow for thermal movement of the pipe

(b) Allow for thermal movement of the wall

(c) Allow settlement of the building

(d) Make maintenance easier

(24) A **dhw** system in which the water drawn off from the taps has passed through the boiler is known as a:

(a) Primary system

(b) Single feed system

(c) Direct system

(d) indirect system

(25) Primary circulation is the water which circulates:

(a) Between the cold feed cistern and the **dhw** cylinder

(b) Between the hot storage vessel and the boiler

(c) Between the hot storage vessel and the furthest draw off point

(d) Around inside the hot water cylinder

(26) To prevent foul water drains surcharging during heavy rainfall, which of the following systems of drainage should be chosen?

(a) Combined

(b) Separate

(c) Relief

(d) Partially separate

(27) Bends are often used on pipes rather than elbows because

(a) The frictional resistance is reduced

(b) The frictional resistance is increased

(c) The pipe work is easier to run

(d) Insulation material is easier to apply

(28) Primary circulation is the water which circulates

(a) Between the cold feed cistern and the dhw cylinder

(b) Between the hot storage vessel and the boiler

(c) Between the hot storage vessel and the furthest draw-off point

(d) Around inside the hot water cylinder

(29) The followings are part of a drainage system except?

(a) Inspection chamber

(b) Pipes

(c) Septic Tank

(d) Plunger

(30) A cesspool is used for which of the following?

(a) To drain out foul water

(b) To store foul water

(c) To store wastes from the building for future use

(d) To remove effluent out of the building.



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**UNIVERSITY OF ILORIN**

**FACULTY OF ENGINEERING AND TECHNOLOGY**

**THIRD SEMESTER EXAMINATIONS 2017/2018 SESSION**

**ABE 222/BME222/CHE 222/CVE 222/CPE 222/ELE 222/MEE 222/FBE 222/MME 222/WRE 222**

**Course unit: 6 Credits** **Time allowed: 2 h**

**INSTRUCTION: Answer all questions carefully**

**SECTION A:** CENTRAL WORKSHOP

## GLASS BLOWING SECTION

**Fill in the gaps or choose from the options provided**

1. Which of the following is a glass former (a) PbO (b) $Fe_2O_3$ (c) $SiO_2$ (d) $NaO_2$
2. ..... is a fluxing agent (a) CaO (b) PbO (c) $Na_2O$ (d) $B_2O_3$
3. ..... is used for cutting cylindrical glasses (a) Cylindrical knife (b) Glass cutter (c) Glass knife (d) Cutter

The three skills in glass blowing are: 4. Cutting 5. rotational and 6. drawing

7. .....is used as "bond strengthener" (a) $SiO_2$ (b) $B_2O_3$ (C) $Na_2O$ (D) CaO
8. .....is the skill in glass blowing used to maintain the shape of the glass in the flame. (a) Rolling skill (b) Revolution skill (c) Rotational skill (d) pulling skill
9. The most common type of glass is.....(a) Soda lime borosilicate glass (b) Soda lime silicate glass (c) Diamond glass (d) Lead oxide glass
10. concentration.....is the most important safety precaution to be observed in a glass blowing workshop.

## ELECTRICAL MAINTENANCE SECTION

13. An arrangement of cable in the distribution is called ……
    (a) Contact (b) Connector (c) Circuit (d) Controller
14. The fuse is connected across the ..load..... melt off when there is an excess voltage.
15. The meaning of M.C.B. is …… (a) More control boards (b) Miniature circuit breakers (c) Many conductor breakers
16. Overload relay is connected to ……..(a) Remote (b) Control switch (c) Contactor
17. A 3- Phase motors is connected with a neutral in other to serve our ……

(a) Sockets (b) Control board (c) Indicating lamps (d) Florescent tubes

18. ……10mm…… is used for service cable
19. The bottle fuse is found in the ……(a) Electrical wire (b) Electrical panel (c) Junction box
20. In the three-phase connection, how many numbers of connected wire do we have;
    (a) 2 wire connected (b) 8 wire connected (c) 4 wire connected
21. A switch is connected to break the following: (a) Socket (b) Overload (c) Circuit (d) Commercial
22. Resistor is one the components that performs the flowing except: (a) Resist current (b) Allows current to pass (c) Heat when excess current flows (d) Pass current to other components



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## WOOD WORK SECTION

23. (a) ...Hack saw..... is suitable for cutting accurately on work piece

(b) Which of the following tools are not a marking out tool (a) G. clamp (b) Try Square (c) Block mile

(c) Loose clothing is most suitable while working on circular machine. True or False

24. (a)...Clamp..... is used for drawing two pieces of wood together when assembling or gluing work

(b) mortise guage is used for cutting wood True or False

c. Mortise chisel is used for sand papering. True or False.

## MACHINING SECTION

25. CNC Machine has the following advantages except.....(a) It is used for mass production

(b) It saves time (c) It is very accurate (d) It is very slow.

26. What is the point angle of a twist drill? (a) $115^0$ (b) $116^0$ (c) $117^0$ (d) $118^0$

27. Geer cutting operation can be performed on.....(a) Lathe machine (b) Milling machine (c) Drilling machine

28. Lathe tail stock can be used to carry out.....(a) Boring operation (b) Drilling operation (c) Taper turning

29. CNC machine programme can be written with...(a) M & Q Code (b) M & B Code (c) M & G Code (d) G & C Code

30. Lathe Machine has the following parts except.....(a) Cross slide (b) Head stock (c) Compound slide (d) Knee

31. The following machines can be classified under machine tool except...(a) Lathe Machine (b) Drilling Machine (c) Electrical power saw

32. Turning tools are made from various materials except.....(a) tools steel (b) High speed steel

(c) Dead mild steel (d) Alloy tool steel

33. Facing operation can be carried on.....(a) Surface grinding machine (b) Shaping machine

(c) Lathe machine (d) Drilling machine

34. The capacity or a lathe machine can be determined by.....(a) The largest diameter of a job you hold in the four-jaw chuck

(b) The largest diameter of a hole you can bore on the lathe (c) The longest length of a bar you can turn between the two Centers.

## FITTING SECTION

35. Drills are ..... tools fixed in drilling machine to make hole (a) facing (b) cutting (c) Drilling (d) knurling.

36. Dies are used for making threads (a) internal (b) vertical (c) external (d) orricenter

37..... is used for measuring inside, outside, and depth dimension in an object.

(a) File (b) steel rule (c) Vernier caliper (d) Vernier Height gauge

38..... are the hand tools used for removing or reducing the size of a metal. (a) G. Clamp (a) Try square (c) Hack saw (d) files.

39..... is used for holding a work piece while filling. (a) Anvil (b) Table vice (c) Center punch (d) Drilling machine.

40..... Machine is used for cutting material to a required length (a) Tin cutter (b) power saw (c) anvil (d) scriber

41. Ball pen hammer is used for ..... (a) Punching (b) dressing (c) cleaning (d) scraping.



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42. The following are precision measuring instrument except .... (a) Steel rule (b) Vernier caliper (c) Vernier height gauge (d) micro-meter

43. Magnetic V. block is used for ..... (a) Holding a work piece (b) filling (c) cutting (d) scraping.

43..... is used for holding short work piece to the bench. (a) G. clamp (b) Try Square (c) scriber (d) Steel rule.

**SECTION B:** EQUIPMENT MAINTENANCE

**INSTRUCTION:** Please fill in your answers on this sheet and submit

Dept.:----------Matric No: ----------

1. Instrumentation entails ---------- and control process variables within the laboratory, working environment, factories etc.
2. The ultimate concern of the instrument process involves choice of right monitoring devices and it is based on;

   i. cost ii.-------- iii.---------- iv. ---------- and, v----------
3. Maintenance is subdivided into basic types as ----------, ----------, and----------
4. Taking measurement of equipment is important due to the following three reasons:

   i. ii. and iii.
5. Troubleshooting implies using the skills acquired from----------, ---------- and ---------- to bring about the localization of a faulty symptom and correcting it to meet the standard of performance requirement.
6. ---------- and ---------- are the transformation variables in a practical power transformer.
7. A component marked as "D12" conducts only in forward bias and does not in reverse direction, what is the component? ----------
8. If the component marked "D12" conducts in both directions what is the condition of such? --------
9. The ---------- is a component often found in the filter circuit of the power supply unit and the values are specified in ---------- and ----------
10. In a computer system the Mother Board is not same as the Central Processing Unit because of ----------, ----------, and ----------
11. In practice a 0 Hz, 22 $V_{dc}$ can be obtained from a 50 Hz 200$V_{ac}$ through the process of ----------
12. If a pressing Iron fails to regulate what component would you suspect? ----------
13. In practice, dust is evacuated from an equipment through the use of a ----------
14. In equipment troubleshooting, resistance is measured using ----------
15. What voltage value would be obtained from LM7805 regulator? ----------



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**SECTION C:** CIVIL ENGINEERING **(Bricklaying & Plumbing Unit)**

**INSTRUCTION:** Answer All

1. .....is the term used for keeping the concrete well all desired purpose of ensuring complete hydration of cement. (curing)
2. .....is the application of mortar on the external surface of walls and ceilings which provide flat even surface.
3. .....is made up of cement, fine aggregate, coarse aggregate, water and reinforcement.
4. .....is the portion of a brick or block cut across the width or cut by some fraction of its length.
5. .....is the correct measurement of the various materials used by concrete mix.
6. Concrete is poor generally good in .....and ..... but good in
7. .....are added to the concrete primary to gain the tensile strength in the concrete.
8. .....is the swelling out of any portion of a wall, due to the presence of oil on the surface of the block before plastering
9. .....is caused by the presence of salt in any of the combined materials, which eventually lead to white appearance on the surface of the wall.
10. The two major type of concrete are .....and.....
11. .....is the stepped arrangement formed during the construction of a wall when one portion is built to a greater height than that adjoining.
12. .....is the material that consist of both fine and coarse aggregate.
13. The proportion of any concrete could be influenced by the ....., ....., .....and.....
14. Low heat cement is used for mass concrete work as dams, bridges True or False
15. Sulphate resisting cement is recommended for the construction of docks & harbours on a soil where sulphate us present TRUE or FALSE
16. A mortar placed in between structural units provides a bond and take part in bearing the load TRUE or FALSE
17. Better concrete work does not depend on better calculations, it depends equally only on better workmanship (TRUE or FALSE)
18. The type of cement which is most suitable for the repair of potholes on road works is rapid hardening portland cement TRUE or FALSE
19. 1:1:6 of gauged mortar is suitable for most conditions of severe exposure. TRUE or FALSE
20. 1:3 of lime mortar is generally good for most conditions of severe exposure TRUE or FALSE
21. 1:2:6 of compo mortar is good for most conditions of severe exposure TRUE or FALSE
22. Cement mortar is not suitable for works in exposure condition TRUE or FALSE
23. 1:3:12 of gauged mortar is good for internal works only TRUE or FALSE



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**SECTION D:** AGRICULTURAL AND BIOSYSTEM ENGINEERING

**INSRUCTION:** ANSWER ALL THE QUESTIONS

~~AUTOMOBILE UNIT (DIESEL ENGINE)~~ VEHICLE MAINTENANCE

1. …….. Is the primary mover of an engine? (a) Flywheel (b) Piston (c) Crankshaft (d) Camshaft
2. TRUE OR FALSE

The heavier a vehicle is, the more fuel it will consume

3. TRUE OR FALSE

You can conveniently run your automobile with a good battery but a defective Alternator

4. The power generated in an engine resides where?

(a) in the engine (b) at the flywheel (c) at the wheels (d) none of the above

5. …….. must generate its own power and transport from point A to B

A, Transportation B, An Aircraft (c) A vehicle (d) A car

6. TRUE OR FALSE

There is a direct correlation between the Electrical ignition system of a vehicle and the Electric arc welding principle.

7. …….. Is necessary for speed variation in a vehicle

(a) Engine (b) gearbox (c) speedometer (d) clutch disc

8. What does ABS stand for?

(a) Autonomous brake system (b) Automatic brake system (c) Antiskid brake system (d) none of the above

9. The limited slip differential is often found in which type of vehicle drive system?

(a) Rear wheel drive system (b) All wheel drive system (c) Front wheel drive system (d) All of the above

10. …… is a type of manual gear transmission

(a) Sliding mesh gear (b) Constant mesh gear (c) synchronous gear (d) All of the above

11. TRUE OR FALSE

Front wheeled vehicles have better weight distribution than Rear wheeled vehicles.

12. The power that is generated by the piston during the power stroke of a 4 stroke ICE is known as?

(a) BHP (b) ABS (c) IHP (d) AEB

13. TRUE OR FALSE

The Indicated Horse Power (IHP) of a vehicle is measured at the wheels as compared to the brake horse power measured at the point of generation

14. The primary source of electrical energy in a vehicle is?

(a) Alternator (b) Battery (c) Starter motor (d) Ignition coil

15. One of the following cannot be classified as a vehicle

(a) Human being (b) horse (c) none of the options (d) all of the options



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SECTION E: MECHANICAL ENGINEERING

## REFRIGERATION AND AIR-CONDITIONING

1. A refrigeration system
   A. Removes heat from a low temperature body and delivers it to a high temperature body
   B. Removes heat from a high temperature body and delivers it to a low temperature body
   C. Rejects energy to a low temperature body
   D. Absorbs energy to a low temperature body
2. During refrigeration cycle, heat is rejected through the
   A. Compressor B. Evaporator C. Condenser D. Leaking pipes
3. The conditioned air supplied to the room must have the capacity to take up
   A. Room Sensible heat only B. Room Latent heat only C. A and B D. None of the above
4. What is the process of heat absorption from the refrigerated space to cause cooling?
   A. Sublimation B. Air Conditioner C. Evaporation D. Refrigerator
5. In a refrigerating machine, heat rejected is …….. heat absorbed
   A. More than B. Equal to C. Less than D. Deviant from
6. Defrosting of a refrigerator may be done by stopping the compressor for a short period. **True or False**
7. The name of the refrigerant used to replace R12 is……. A. Dichlorodifluoro methane
   B. Monochlorodifluoro methane C. 1,1,1,2 tetra fluoro ethane D. Dicloromonofluoro methane
8. The condition of refrigerant after leaving the compressor and before entering the condenser is
   A. Liquefied gas B. Superheated Vapour C. Algid liquid D. High pressured and low temperature vapour
9. The device used to control the rate of admission of refrigerant entering into the evaporator is
   A. Contraction device B. Automatic valve C. Metering device D. Power Switch
10. Which of these is not a property of an ideal refrigerant? A. Low saturation pressure
    B. High critical temperature and pressure C. Low specific volume D. High heat of vaporization
11. What is the principle of refrigeration? A. Newton's second law of motion
    B. Third law of thermodynamics C. Law of conservation of energy D. Second law of thermodynamics
12. Which of the following ways cannot be used to detect a leaking pipe in a refrigerator?
    A. Use of halide torch B. Dipping the whole refrigerator in a soapy solution
    C. Fluorescent leak detection method. D. Electronic leak detection
13. Which of these is not a type of water-cooled condenser?
    A. Double pipe B. Tube and Shell C. Shell and Coil D. None of the above
14. After detecting points of leakage, the pipes must be …… with filler metals
    A. Brazed B. Soldered C. Welded D. Replaced
15. If halide torch method is to be used for leakage detection, the refrigerant must not be of
    A. Low saturation pressure B. High freezing point C. Flammable D. Low critical temperature and pressure



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(1) EGR is a device that measures.....

(a) The flow of water (b) The flow of air (c) Captured uncombusted

(2) Zirconia-dioxide is a sensor property that is used to measure.....

(a) Air flow (b) Carbon monoxide (c) Oxygen

(3) What is the full meaning of the acronym for "ECU"?

(a) Exhaust Combustion Unit (b) Electronic Compartment Unit (c) Electronic Computer Unit.

(4) Which of the following is an actuator?

(a) Steeper Motors (b) Piezoelectric (c) Thermistor

(5) Which of the following is a resistive sensor? (a) Thermistor (b) Solenoids (c) Speed Sensor

(6) Oxygen Sensor is located in the .... (a) Engine (b) in the Catalyst (c) Exhaust Manifold

(7) Which of the following is correct about Negative temperature coefficient?

(a) The lower the resistant higher the temperature (b)The higher the temperature the lower the resistance

(c) The lower the temperature the lower resistance

(8) Change in the output of sensor with change in in input is;.....(a) Threshold (b) Resistivity (c) Sensitivity

(9) Applications of light sensor include..... (a) Spectrophotometer (b) Speed Sensor (c) All of mentioned

(10) What is the 'Rich' status of fuel ratio in petrol engine?

(a) 14.7:1 (b)14.6:1 (c)14.7:2

(11) What is the industrial application of load cell?

(a) To measure the flow of water (b) To measure differential weight (c) To test the out flow of liquid

(12) Transducer is a device that Converts.....(a) Energy from middle level to higher level

(b) Energy from one form to another (c) Conserve Energy

(13) Sensor is a type of transducer (a)True (b) False

(14) Thermocouple generate output voltage according to.....(a) Humidity (b) Temperature (c)Voltage

(15) How many types of transducers there are? (a) 2 (b) 4 (c) 7

## VEHICLE MAINTENANCE SECTION

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**SECTION F:** ELECTRICAL AND ELECTRONIC ENGINEERING

1. The following are the conditions for Synchronizing distribution transformers except.....
   a. The transformers must have the same frequency of operation.
   b. They must have the same kVA rating.
   c. They must have the same transformer oil types.
   d. The must have the same voltage rating.

2. The protective fuse type used in a feeder pillar is known as.....
   a. J & P fuses
   b. J & D fuses
   c. HRP fuses
   d. HRC fuses

3. Fourteen numbers of 13A socket outlet will consume..... total current at the distribution board, considering IEEE regulation on final sub-circuits.
   a. 2990A
   b. 60A
   c. 67.5A
   d. 45A

4. The following are the signs of open circuit fault in an electrical installation except.....
   a. Appearance of smoke
   b. Melting of cables
   c. Fire outbreak
   d. None of the above

5. ..... is used in protecting distribution transformer against erosion.
   a. Plinsh
   b. Transformer basement
   c. Guiding wires
   d. Plinth

6. Primary – Secondary wiring configuration of transmission transformer is always.....
   a. Delta – Delta configuration
   b. Star – Delta configuration
   c. Star – Star configuration
   d. Delta – Star configuration

7. The IEEE recommended cable size for electric cookers is.....
   a. $16mm^2$
   b. $10mm^2$
   c. $2.5mm^2$
   d. $4mm^2$



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8. A 2-bed room flat with total current demand of 155.42A - 10 ways at the DB, has future load estimate of 62.58A – 4 ways, ..... distribution board capacity will be appropriate for the building.
a. 200A – 6 ways
b. 250A – 12 ways
c. 300A – 18 ways
d. 350A – 24 ways

9. One of the expressions below explains the installation of a point of light:
a. DB neutral cable to load Neutral cable, DB Live cable to load live cable, and load live cable to switch neutral cable.
b. DB live cable to load live cable, DB neutral cable to switch neutral cable, and load neutral cable to switch live cable.
c. DB neutral cable to load neutral cable, DB live cable to switch live cable, and load live cable to switch neutral cable.
d. None of the above

10. The following are the consequences of wrong fusing except.....
a. Generation of excessive heat at the fuse
b. Burning of fuse coverings
c. High cost of replacing the fuse
d. Burning of installation cables

11. A hand drilling tool used to create hole for easy passage of screw is.....
a. Jimat
b. Jimlet
c. Jimtet
d. Jimget

12. What makes copper conductor preferred to aluminum conductor are.....and .....
a. Conductivity and acceptability
b. Conductivity and ductility
c. Communicability and ductile
d. Conductivity and cost of material

13. The device that determines how current flows into the circuit is called.....
a. Flower
b. Controller
c. Control
d. Stopper

14. The recommended cable size for ceiling fan installation is.....
a. $15mm^2$
b. $1.55mm^2$
c. $0.15mm^2$
d. $1.5mm^2$

15. The plate that contains information about an appliance or equipment is called.....
a. Plate name
b. Plant name
c. Name plate
d. Name plant

16. A resistor connected in series in an electronic circuit is .....in the circuit.
a. To increase the power
b. To reduce the power
c. To limit the current
d. To limit the voltage

17. The power rating of colour coded resistor is determined by.....
a. The colour
b. The size
c. The tolerance
d. The length

18. The capacitor connected in a power supply unit is to.....
a. Remove residual voltage
b. Remove residual current
c. Remove ripple voltage
d. Remove ripple current



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19. The regulator unit is to.....
a. Keep the power constant
b. Keep the circuit constant
c. Keep the current constant
d. Keep the voltage constant

20. The filtering section of a power supply unit comprises of..... and .....
a. Transistor and diode
b. Resistor and diode
c. Resistor and transistor
d. Capacitor and inductor

21. The difference between the fixed and variable resistors is.....
a. To provide variable power
b. To provide variable value
c. To provide variable potential
d. To provide variable color

22. Power supply unit comprises of..... sections.
a. Six basic sections
b. Two basic sections
c. Four basic sections
d. Five basic sections

23. The basic function of a transistor is.....
a. Current / voltage Amplification
b. Current / voltage Regulation
c. Current / voltage Reduction
d. Current / voltage Induction

24. The two basic techniques of rectification are..... and .....
a. Constant / low wave Rectification
b. Pure / Smooth wave Rectification
c. Reduction / Induction wave Rectification
d. Half / Full wave Rectification

25. The transformer in a power supply unit is to.....
a. Raise up the voltage
b. Step down the voltage
c. Convert the voltage
d. Smooth the voltage

26. Soldering is a method of.....
a. Joining metals together
b. Joining PCB together
c. Joining components together
d. Joining circuit together

27. A good quality soldering must be.....
a. Qualitatively and Rigidly sound
b. Mechanically and Electrically sound
c. Cohesively and Adhesively sound
d. Smoothly and Brightly sound

28. The three sequential procedures of achieving a good soldering are.....
a. Warming, Cleaning and Smoothing
b. Wiring, Cleaning and Brushing
c. Cleaning, Tinning and Heating
d. Heating, Cleaning and Joining

29. The hottest part of the soldering iron is called.....
a. The filament
b. Heating Element
c. The hot plate
d. The iron bit

30. The type of soldering adopted in electronics is known as _____
a. Nice Soldering
b. Soft Soldering
c. Good Soldering
d. Hard Soldering



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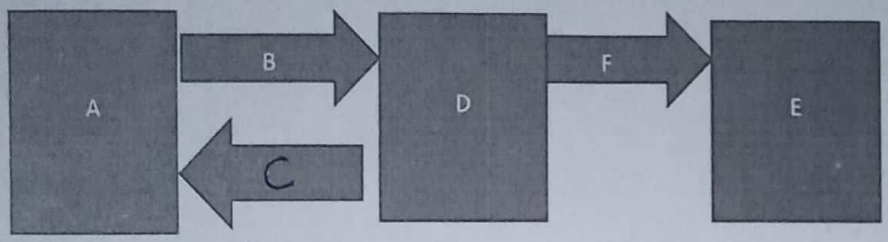
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Complete the Table below

| S/N | Heart Condition | BPM | Respiratory rate |
|---|---|---|---|
| 18. | Normal | 60 | 20 |
| 19. | Bradycardia | 30 | 15 |
| 20. | Hypertension | 135 | 500 |
| 21. | Tachycardia | 192 | 50 |
| 22. | Hypotension | 40 | 15 |
| 23. | Asystole | 0 | 0 |

24. The strength of contraction of the heart are modified by the ....Sympathetic..... and ....para-sympathetic... division of the autonomic nervous system.
25. Calculate the pulse pressure for a blood pressure measurement of 136mmHg/80mmHg.
26. The heart condition in which R-R interval of the ECG waveform of the heart is wider than normal are .....Brady.....cardia..... and .....hypertension.....
27. The flat line on ECG waveform signifies .....Asystole.....

Complete this biomedical instrumentation diagram with the appropriate words




28. A. .....~~measurand~~ measurand..
    B. .....actuator.....
29. C. .....sensor.....
    D. electric instrumentation/signal control
30. E. .....output display bay storage transmission
    F. .....signal processing.....



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